

YASIR RAHMAN

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About Me

I'm a **Software Engineer** and **Researcher** with interests in **security vulnerability analysis** and **trustworthy systems**, focused on analyzing vulnerabilities in complex systems and developing solutions to build **reliable and trustworthy systems**. I aim to pursue a **PhD** in this area to bridge research with practical impact.

Education

Chittagong University of Engineering and Technology
BSc in Computer Science and Engineering

Feb, 2020 - Jun, 2025
CGPA: 3.53 / 4.00

- Awarded **merit scholarships** in all **8 semesters** and represented CUET in **ICPC Dhaka Regional Contests**.

Test Scores

GRE: **324 / 340** (Quantitative: **169**, Verbal: **155**, Analytical Writing: **4.0 / 6.0**)

Research Experience

Memory Injection Attacks on Long-Term Memory of LLM Agents *July 2026 - Present*
Ongoing Research *In collaboration with Dr. Shahnewaz Karim Sakib and Dr. Anindya Bijoy Das*

- Investigating existing **memory injection attacks** on LLM agents' long-term memory and their limitations.
- Analyzing LLM agent memory vulnerabilities to improve both attack strategies and defense mechanisms.

Proactive Security for Cross-Chain Bridges Against Adversarial Attacks *July 2026 - Present*
Ongoing Research *In collaboration with Dr. Shahnewaz Karim Sakib and Dr. Anindya Bijoy Das*

- Investigating how adversaries, including passive attackers, exploit vulnerabilities in cross-chain bridges and transactions.
- Identifying weaknesses in bridge architectures and developing **proactive security mechanisms** against them.

Attack Scenarios in Cross-Chain Blockchain Interactions *January 2026 - Present*
Ongoing Research *In collaboration with Dr. Rahma Bintey Mufiz Mukta*

- Conducted a comprehensive survey and taxonomy of security vulnerabilities and attack patterns in cross-chain communication protocols.
- Collected and analyzed **50+ real-world cross-chain bridge attack incidents** from 2022 to August 2026, drawing from sources such as bridges' own post-mortem reports and third-party security audits.
- Mapped attacks to specific attack patterns and functional domains based on analysis.

Predictive Model for Birth Delivery Mode Using EHG Signals *September 2023 - November 2025*
Research Assistant (Freelance) *In collaboration with Dr. Rubana Hoque Chowdhury*

GitHub Repository: <https://github.com/ChyYasir/Predictive-Model-for-Birth-Delivery-Mode>

- Designed and implemented **machine learning pipelines in Python** for predicting birth delivery mode from electrohysterogram (EHG) signals, covering signal preprocessing, feature extraction, model training, and evaluation.
- Re-implemented methods from prior published EHG studies as baselines and ran iterative experiments to refine models.

Bengali Keywords Extraction from Bengali Blog Using Transformer-Based Approach *2024 - 2025*
Undergraduate Thesis Research *Supervised by Dr. Muhammad Ibrahim Khan*

- Developed a comprehensive Bengali blog dataset of **9,012 annotated posts** and proposed a **transformer-based dual-encoder architecture** achieving **84% F1@10 score** and did some comparison analysis with the existing methods.

Publications

- (2026). *Analysis of electrohysterogram signals for predicting obstetric outcome using machine learning methods: a scoping review*. **BMC Pregnancy and Childbirth**, Springer Nature (Q1). ([Link](#))

Teaching Experience

International Islamic University Chittagong (IIUC) *February 2026 - August 2026*
Adjunct Lecturer, Dept. of Computer Science and Engineering *Chittagong, Bangladesh*

- Taught undergraduate courses in **Algorithms** (theory & lab), **Compiler Design** (theory & lab), and **Competitive Programming II** (lab).
- Developed a content website, [Yasir Explains](#), to supplement course material for students.

Industry Experience

Bevy Commerce *July 2024 - Present*
Software Engineer (L3) *Ontario, Canada (Remote)*

- Implemented backend services and frontend features for an **online news portal** serving over **500,000 DAU**.
- Developed a **data migration tool** to move data from **Magento** and **NetSuite** to **Shopify** using **Shopify GraphQL mutations**, successfully migrating over **5 million data points**.
- Built **SFTP**-based file transfer pipelines and automation scripts for client data migrations, cutting down manual intervention and reducing failed transfers.

BreakByte

Software Engineer

December 2023 - June 2024
Chittagong, Bangladesh

- Implemented secure authentication and authorization for a full-stack blogging platform and integrated the **Tiptap** editor for rich content creation.
- Developed a scalable **multi-tenant architecture**, reducing onboarding time by approximately **50%**, and deployed the platform on **AWS EC2/S3** for secure, scalable infrastructure.

Technical Skills

- Languages:** Python, C, C++, JavaScript
- ML Tools and Libraries:** PyTorch, Scikit-learn, Pandas, NumPy, TensorFlow
- Frameworks:** React, Node.js, Express.js, Next.js
- Databases:** MongoDB, PostgreSQL, MySQL, Redis
- Cloud and Tools:** AWS (EC2, S3), Docker, Git

Projects

Rilo EMS (Employee Management System)

Next.js, TypeScript, MongoDB, ShadCN

GitHub Repository: <https://github.com/ChyYasir/rilo-ems>

December 2024 - January 2025

- Led an **8-member development team** to design and ship a full-featured employee management system in **Next.js**, owning the architecture and code-review process that kept the codebase consistent across all 8 contributors.
- Implemented the **backend boilerplate** and **authentication system**, and integrated the core tools, libraries, and **MongoDB** database that the rest of the team built on.

Easy Tuition (Tuition Management System)

MERN Stack, Material UI

GitHub Repository: <https://github.com/ChyYasir/EasyTuition>

Live Demo: <https://easy-tuition.onrender.com/>

October 2023 - June 2024

- Built a **MERN-stack** admin dashboard with a tutor-matching algorithm and optimized search/filtering across **10,000+ student and tutor records**, cutting administrative decision-making time from **3+ hours to 5-10 minutes**.

Awards & Achievements

- ICPC Regionalist, Dhaka Region:** 2021-22 (Rank 48) & 2022-23 (Rank 41)
- East Delta University Engineering Day Programming Contest 2022:** Rank 1
- CUET Inter University Programming Contest 2022:** Rank 10
- BUET Inter University Programming Contest 2022:** Rank 47
- Sylhet Inter University Junior Contest:** Rank 12
- 2nd Runner-up:** CUET Intra Junior Programming Contest 2021 (Rank 3)
- Top 16 Finalist:** BUET CSE FEST 2024 Hackathon (National) | [GitHub](#)
- 2nd Runner-up:** Electrical Surge 2.0 Hackathon (CUET IEEE)
- 2nd Round:** DEVOPS Hackathon, Building Platform Using Cloud (Poridhi, Powered by Brain Station 23)
- Solved 2000+ problems** across online judges: [Codeforces: CHY_YASIR](#), [AtCoder: Chy_chy](#)
- Selected to Represent Bangladesh** at the International Olympiad on Astronomy and Astrophysics (IOAA) 2017

Co-curricular Activities

CUET Computer Club

2022 - 2025

Instructor & DSA Mentor, CP Bootcamp / Competitive Programming Group

Chittagong, Bangladesh

- Mentored students on Data Structures & Algorithms (Graphs, Segment Trees, Bitmask, etc.) for ICPC and university contests. **Recorded Sessions:** [YouTube Channel](#)
- Designed **competitive programming problems** and organized contests for ICPC and IUPC preparation, followed by post-contest discussions.

IEEE Computer Society, CUET Student Branch

2024 - 2025

Vice Chair

Chittagong, Bangladesh

- Organized an intra-CUET microservice hackathon and a competitive programming contest, and conducted multiple educational sessions on software engineering and competitive programming.

Certifications

- ICPC Dhaka Regional Contest:** 2021-22, 2022-23 | [View Certificate](#)

References

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